

AI Express Hackathon: Technical Summary Sheet

Track 1 -- Warehouse Logistics Agent (Unit 2, Informed Search)

1. Header & Team Info

Course Code: BCA301-5

Group ID: Team 02

Team Members: Akash Ganesh 2441604

Aleena Elsa Benoy 2441605

Aman 2441606

Selected Track: Track 1 -- Warehouse Logistics Agent (Unit 2, Informed Search)

<https://github.com/amandalal07/AI-Express-AStar-Grid-Agent>

GitHub Repository:

2. PEAS Framework Matrix

Performance Measure	Deliver the package to the loading bay with minimum path cost (steps); minimise nodes expanded and replanning time.
Environment	2D discrete grid warehouse floor. Static shelf obstacles known upfront, plus one dynamic blockage (a shelf/pallet toppling into an aisle) introduced mid-run -- partially observable and dynamic.
Actuators	Move Up / Down / Left / Right by one grid cell.
Sensors	Full visibility of static shelf obstacles at the start; the forklift only "senses" the new blockage once it appears on its planned route.

3. Core Algorithmic Formulation

State space: every free (non-obstacle) grid cell (x, y) on the warehouse floor. **Initial state:** the forklift's current cell (pickup point at the start). **Goal test:** state == goal (the loading bay). **Actions:** Up / Down / Left / Right, unit cost each. **Path cost:** $g(n)$ = number of moves taken so far (uniform step cost = 1). **Heuristic:** $h(n) = |x - \text{goal}_x| + |y - \text{goal}_y|$ (Manhattan distance -- admissible and consistent on a 4-connected grid, so A* is guaranteed optimal). **Evaluation function:** $f(n) = g(n) + h(n)$. **Dynamic replanning:** when a shelf/pallet topples into the forklift's planned route, A* is re-invoked from the forklift's current cell to the goal, treating the new blockage as an obstacle, and the agent resumes execution on the newly returned optimal path without restarting from the pickup point.

4. Complexity Analysis

Time complexity	$O(b^d)$ worst case; effectively $O(V \log V)$ with a binary-heap priority queue over $V = \text{width} \times \text{height}$ reachable cells	Initial search: 72 nodes expanded, path cost 24, ~0.8 ms. Post-blockage replan: 22 nodes expanded, ~0.2-1.7 ms. Total: 94 nodes expanded across both searches.
Space complexity	$O(V)$ for the open + closed sets	Peak open-set (frontier) size observed: ≤ 12 nodes, on this 160-cell grid ($V=160$)

Observed metrics were captured via `python main.py --headless` and the pygame visual run on the project's fixed default warehouse layout; re-run to regenerate on your own machine / layout before final submission.